STUDENT SUCCESS SIGNALS

A Data Quality & Learning Habits Audit

WeIntern AIML Internship | Week 1 | Business Insight Report

DATA NOTE: This is a simulated practice dataset, not data collected from real students. Results are illustrative and must not be used to make decisions about real individuals.

1. Project Context

This project studies how attendance, study time, assignment scores, and midterm scores relate to final marks. It demonstrates the Week 1 process of data inspection, cleaning, descriptive analysis, visualization, and stakeholder-friendly reporting. No machine-learning model is trained.

2. Objective

- Describe the distribution of final scores.
- Explore associations between final scores and study/engagement indicators.
- Identify and document common data-quality issues.
- Suggest cautious support actions that would require validation with real data.

3. Dataset Summary

Metric	Result
Raw records	1,212
Variables	11
Exact duplicates removed	12
Rows excluded: invalid/missing final score	17
Final cleaned records	1,183
Final score mean / median	76.2 / 77.2 out of 100
Final score standard deviation	11.8

Fields include student ID, gender, age, attendance percentage, study hours per week, assignment score, midterm score, final score, parental education, internet access, and extra activities. The CSV was generated locally for this assignment; no external data source is claimed.

4. Analysis Approach

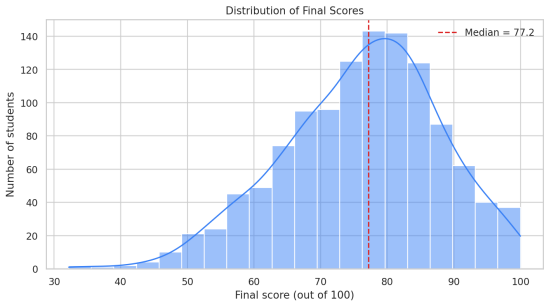
The workflow removes exact duplicates, standardizes category labels, validates numeric ranges, uses median imputation for numeric predictors, keeps missing category values as Unknown, and excludes records without a valid final score rather than fabricating target values. Summary tables and six charts are saved alongside the notebook.

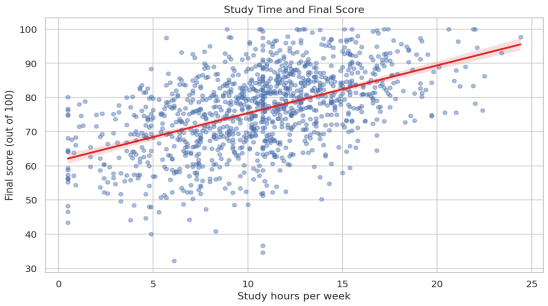
5. Key Findings

The statements below describe the simulated dataset only. Correlations are linear associations, not proof of cause and effect.

1. The cleaned dataset contains 1,183 records. Final scores average 76.2/100 (median 77.2; standard deviation 11.8).
2. Study hours and final score have a Pearson correlation of r = 0.51, a positive linear association in this simulated sample.
3. Attendance and final score have a Pearson correlation of r = 0.43; this is descriptive and does not show that attendance alone causes a score change.
4. Assignment score and final score have a Pearson correlation of r = 0.70.
5. Midterm score and final score have a Pearson correlation of r = 0.74.
6. 23.3% of cleaned records have attendance below 75%; this threshold is an illustrative monitoring flag, not a validated intervention cut-off.
7. 1.7% of cleaned records have final scores below 50/100; these records could be reviewed as a group in a real pilot, with safeguards and context.
8. 1.6% of records have internet access recorded as Unknown after cleaning, indicating a need to improve response completeness.

Visual evidence





The final score mean is 76.2/100. Correlation with study hours is r = 0.51; correlation with attendance is r = 0.43. Assignment and midterm scores are more strongly associated with final score in this synthetic sample, partly reflecting how the data was generated.

6. Recommendations

1. Pilot an early-support review for students with repeated low attendance or falling assessment scores; use human review rather than an automated label.
2. Offer study-planning resources and optional tutoring to students who request help, then evaluate whether the support improves outcomes.
3. Use assignment and midterm performance as discussion prompts for teachers, while considering course difficulty and grading differences.
4. Improve completeness of fields such as internet access and parental education, and document why each field is collected.
5. Validate the patterns on a real, representative dataset and examine differences across groups before adopting thresholds or interventions.

7. Limitations

- The dataset is simulated. Findings cannot be generalized to any real school or student population.
- Correlation does not establish causation, and other factors may contribute to observed patterns.
- Median imputation is a simple demonstration and may affect distributions; track imputed values and test sensitivity in real data.
- Attendance below 75% is only an illustrative flag, not a validated threshold for intervention.
- Any real student dataset requires privacy protection, appropriate consent, limited access, fairness review, and educator oversight.

8. Conclusion

The workflow suggests moderate positive associations between study hours and final score (r = 0.51) and attendance and final score (r = 0.43). Assignment and midterm scores show stronger associations in this simulated sample (r = 0.70 and r = 0.74). These values are mainly useful for demonstrating analysis methods; the primary deliverable is the documented, reproducible workflow. Validate any conclusions on representative real data before taking action.

Project files: `notebooks/week1_analysis.ipynb`, `data/cleaned/`, `outputs/charts/`, and `outputs/tables/`.